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Artificial Atom Lattice

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OUTLINE

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1. Background

2. Device Design

3. Proposed Results

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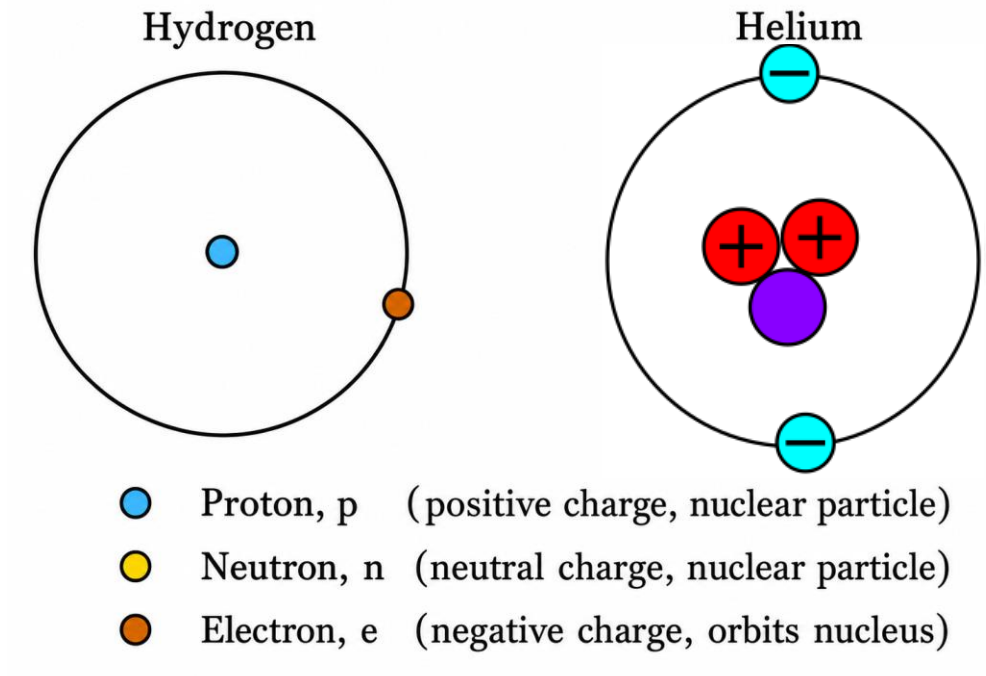
01.

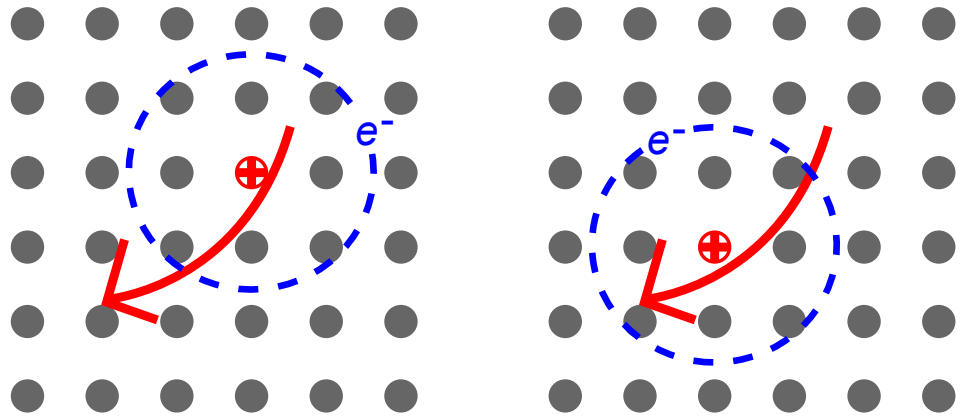
Background

Atoms and excitons

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Can a programmable positive-charge lattice turn graphene into a new many-body material?





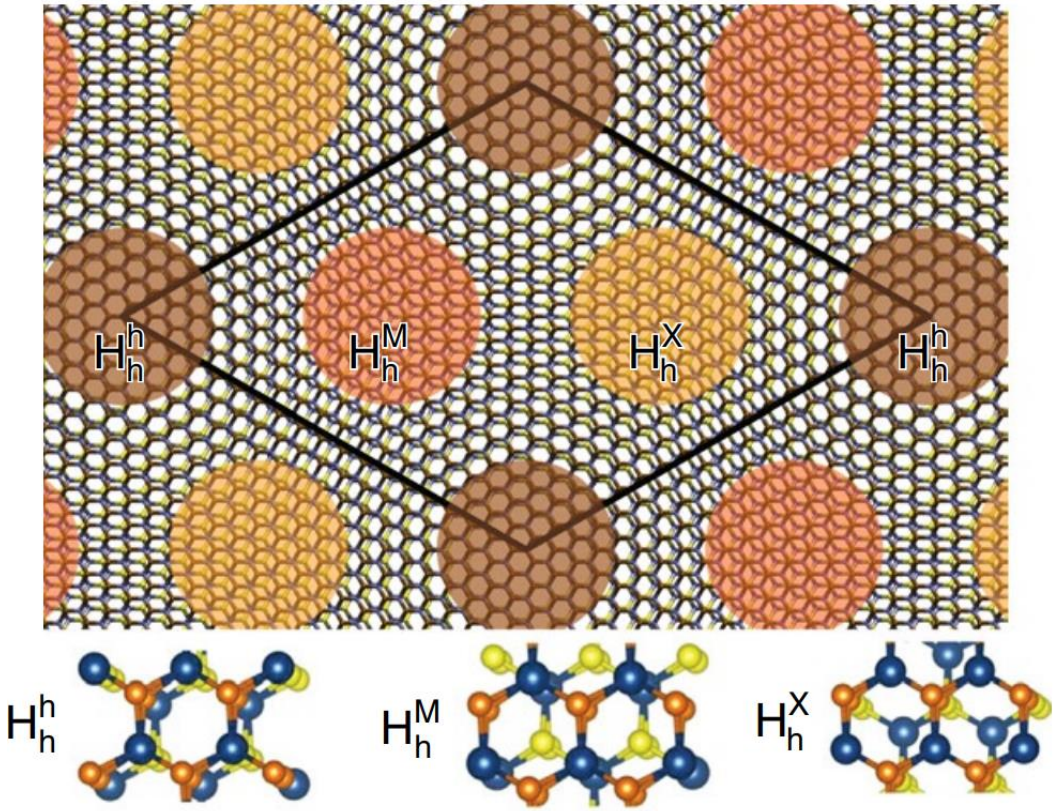
Wannier–Mott exciton

Positronium?

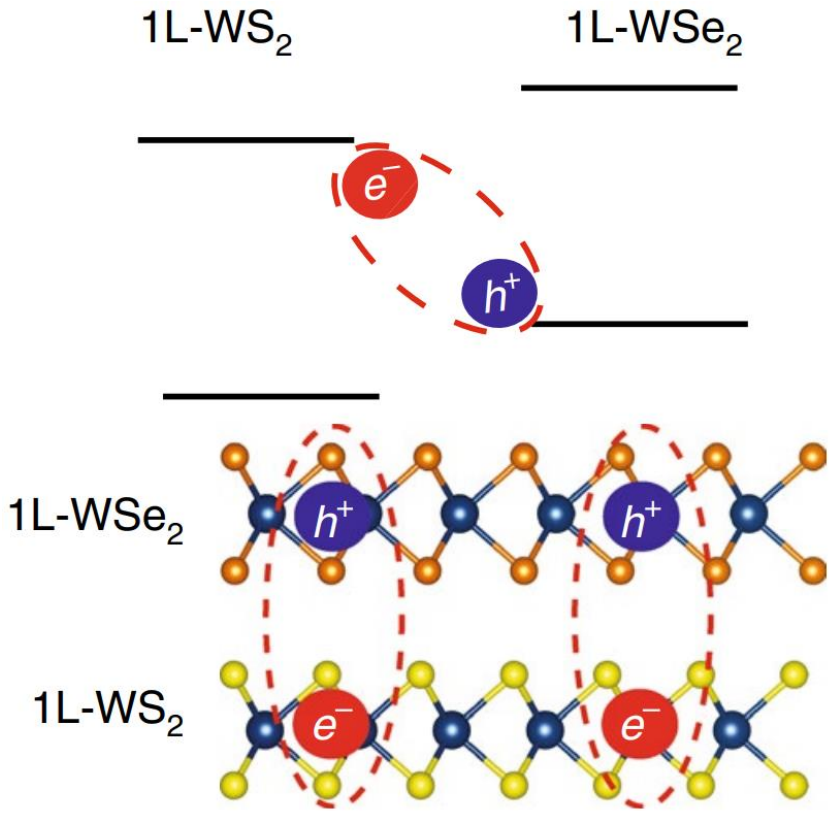
Moiré and interlayer exciton

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Periodic Moiré potential



Type II band alignment



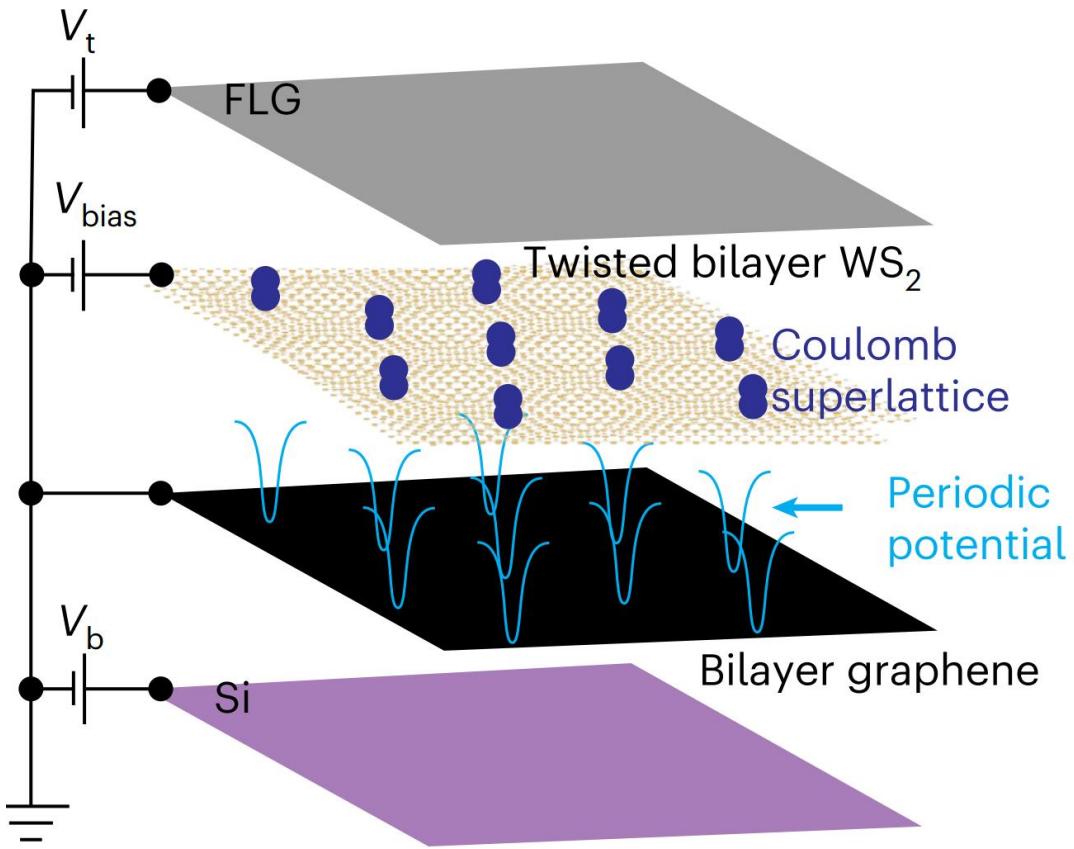
Nature Materials 19, 617–623 (2020)

Bernal bilayer graphene

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Tunable – contact/ available for Hall measurement

Magnetic length (Orbital radius)



Why bilayer graphene?

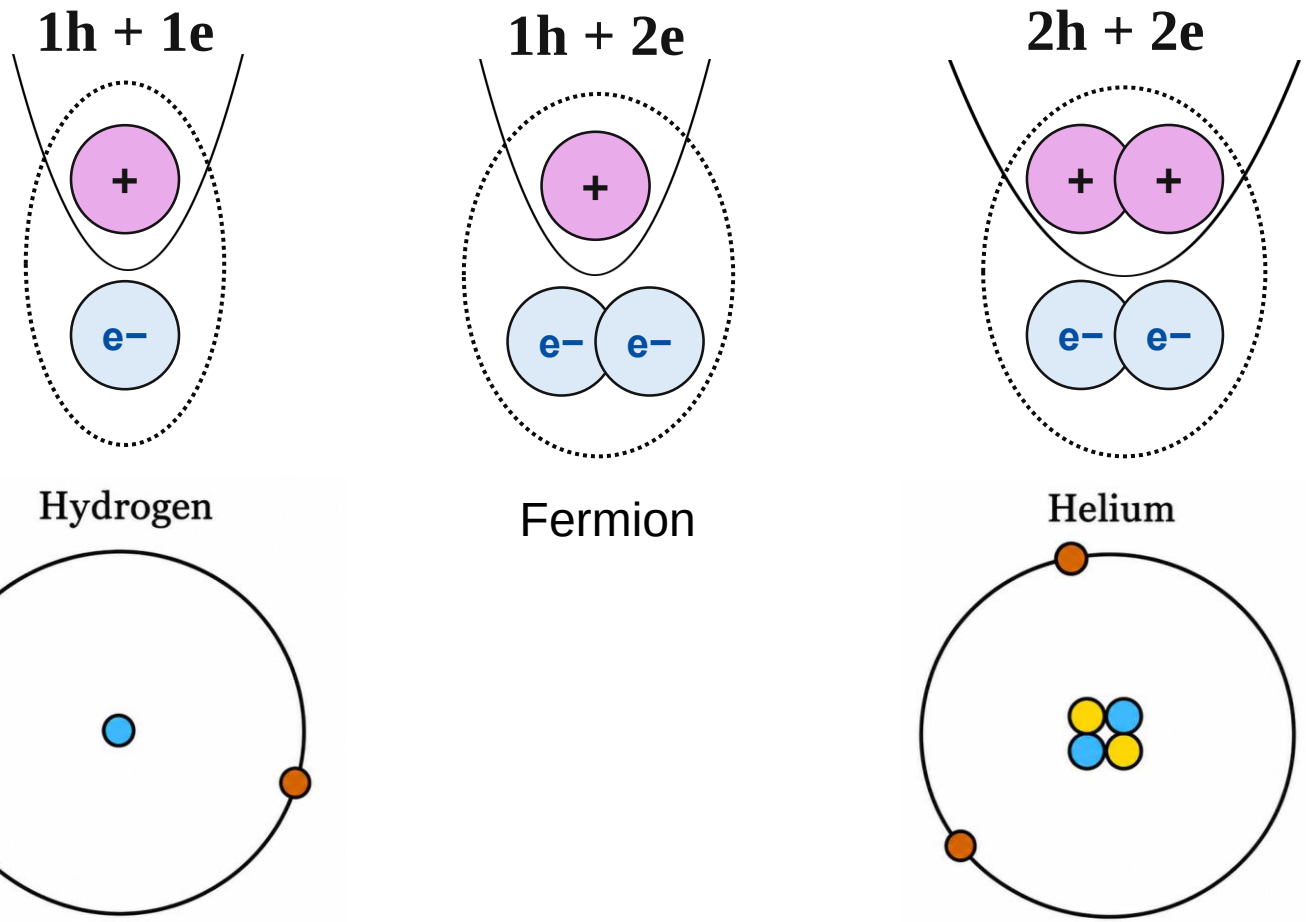
Different kinetic energy

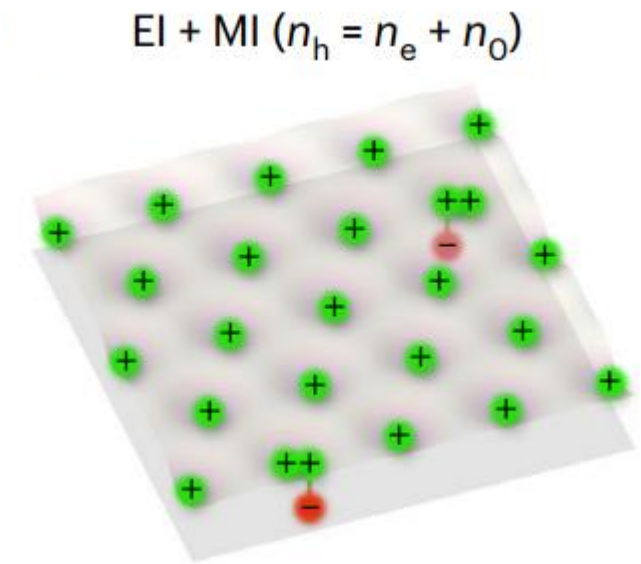
Nature Materials 23, 189–195 (2024); PRX 8, 031023 (2018)

Possible states in trap

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The atomic label is valid only after the center charge and electron occupancy are known.





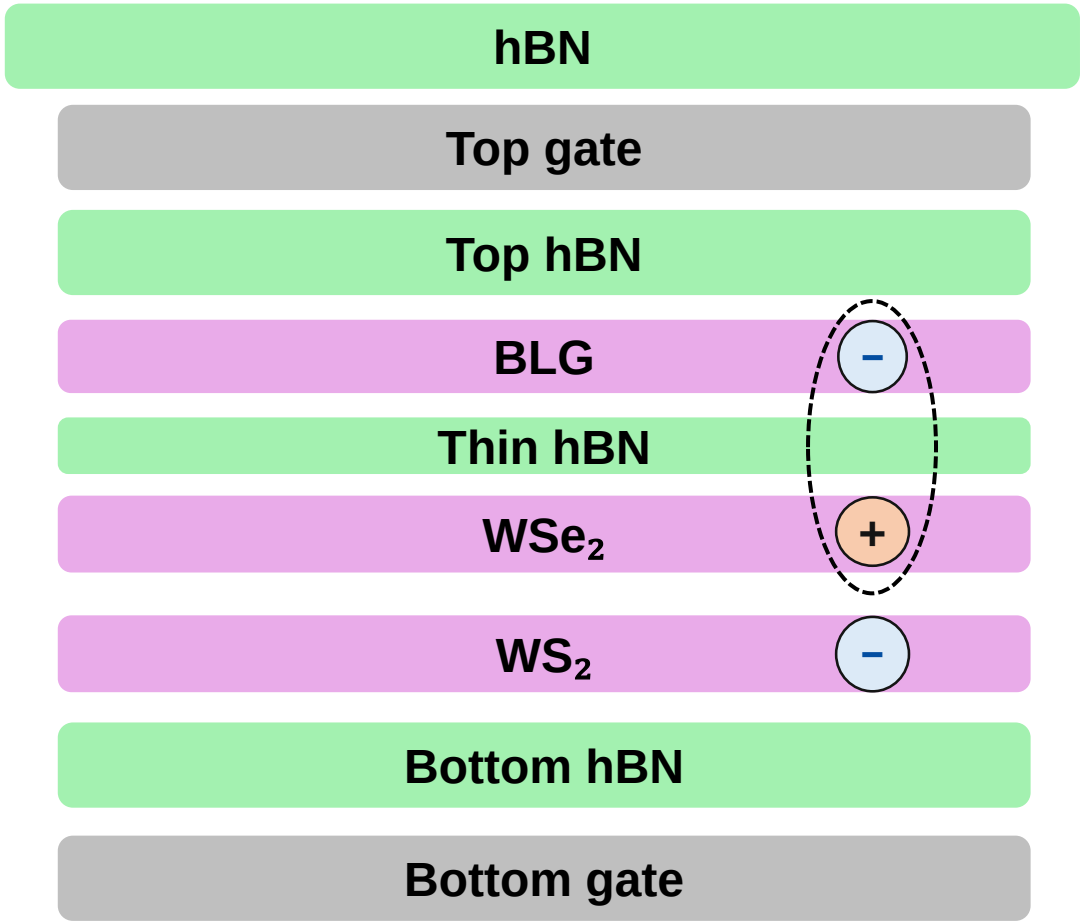
PRL 127, 256802 (2021); Nature Physics 22, 514–520 (2026)

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02.

Device Design

Device structure



$V_{T,C,B}$ Gate Voltage: Top, Contact, bottom

$C_{t,i,b}$ HBN Capacitance: Top, Thin, bottom

$(\phi, u, n)_{e,h}$ Potentials, carrier density: Electron, Hole

Gauss' law

$$-en_e = C_t\left(\phi_e - V_T\right) + C_i\left(\phi_e - \phi_h\right)$$
$$-en_h = C_i\left(\phi_h - \phi_e\right) + C_b\left(\phi_h - V_B\right)$$

Nature Physics 22, 514–520 (2026)

Interesting Properties

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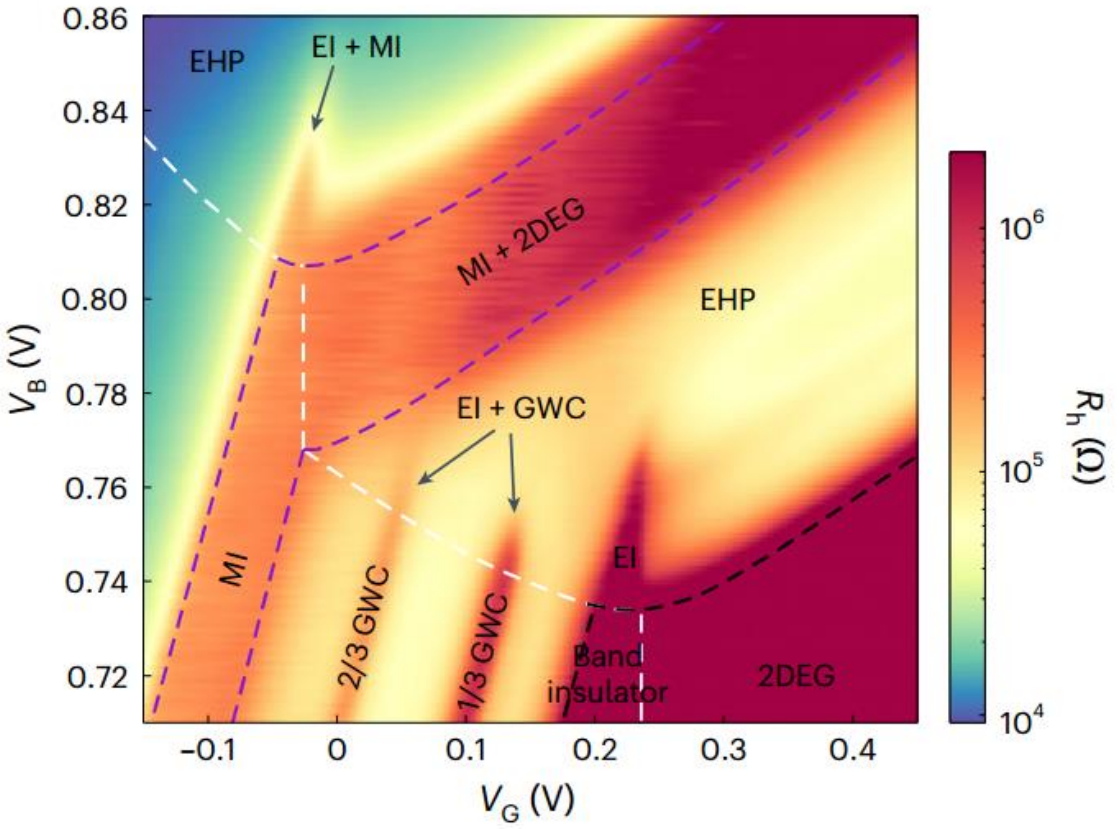
Thermodynamic behavior: isothermal compressibility

Quantum Statistics of Boson and Fermion

BEC/ Fermi liquid
Regard of t/U ratio: Mott insulator, Wigner crystal or correlated lattice fluid (supersolid)

Dynamics of artificial atom lattice system

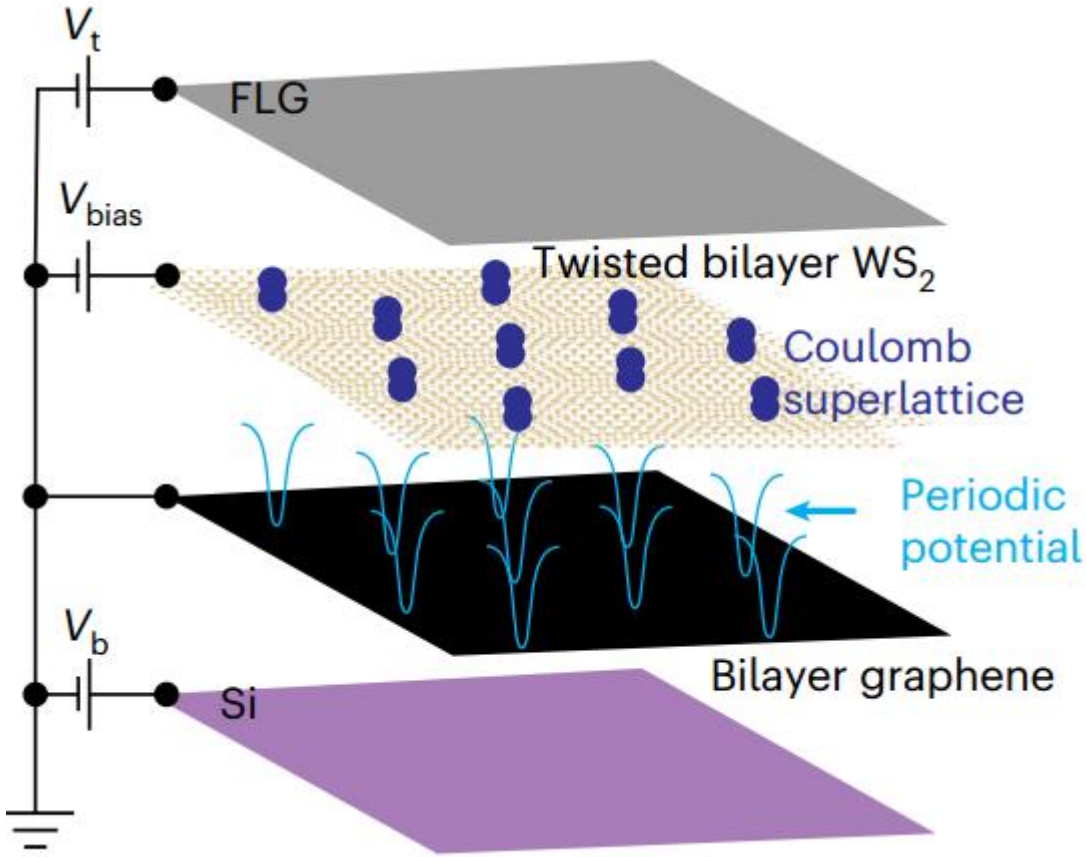
Coulomb drag/ Counter flow/ BKT

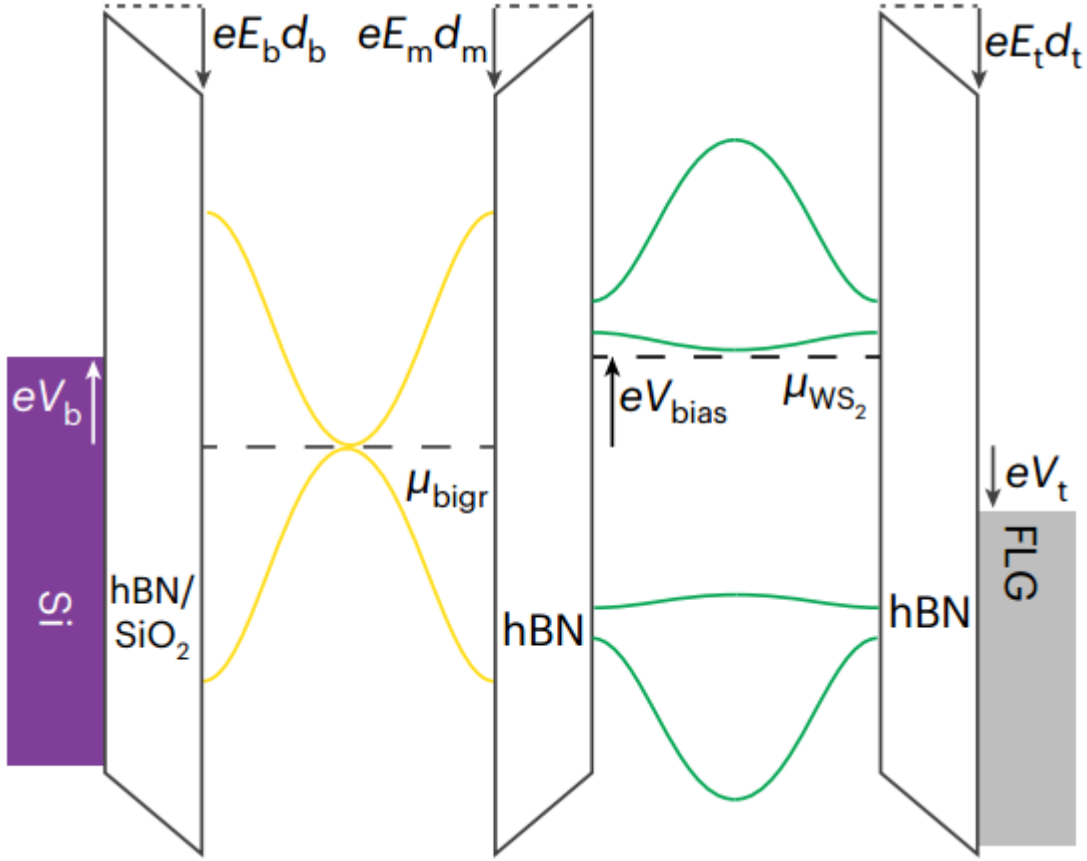


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Related paper

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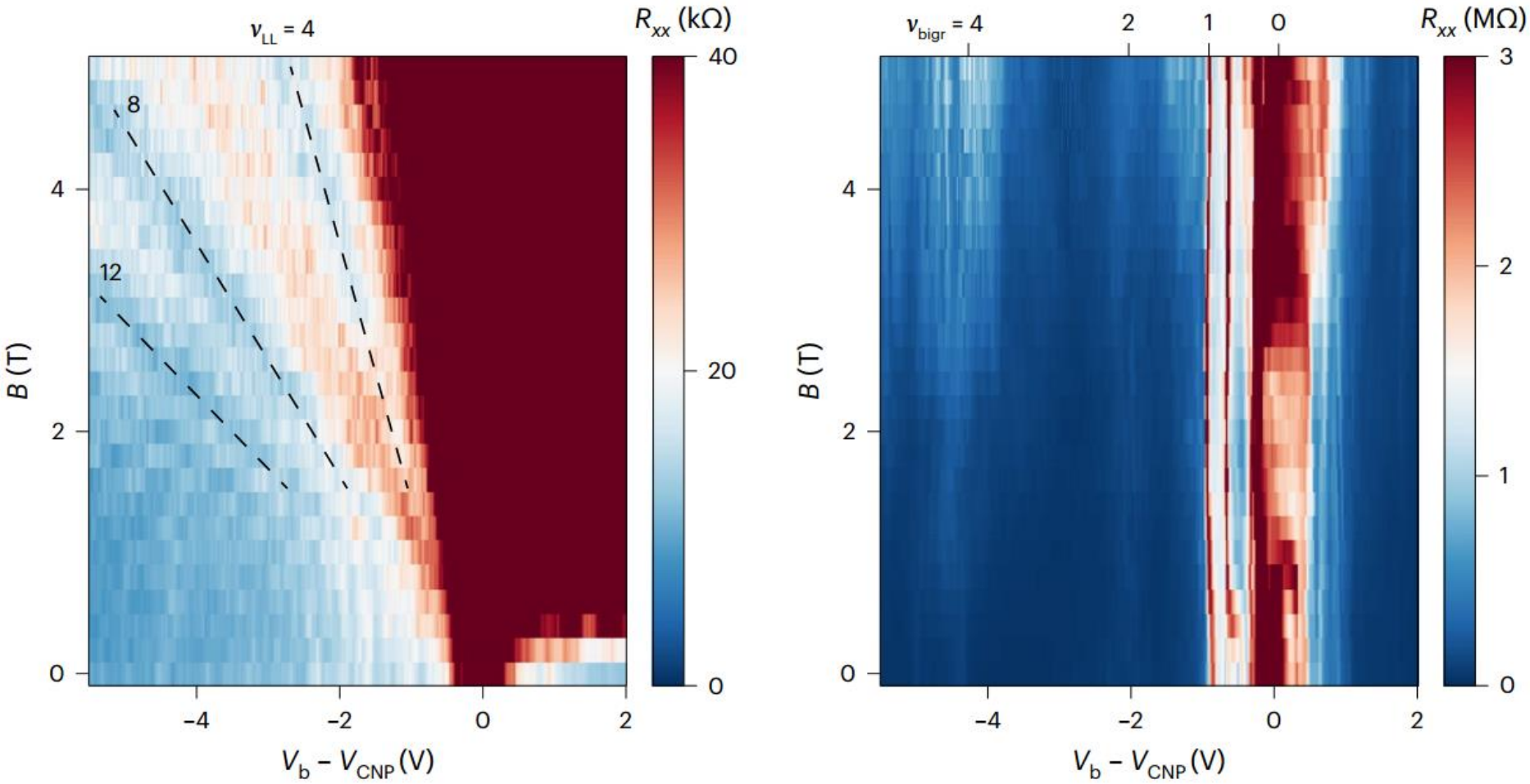




Nature Materials 23, 189–195 (2024)

Related paper

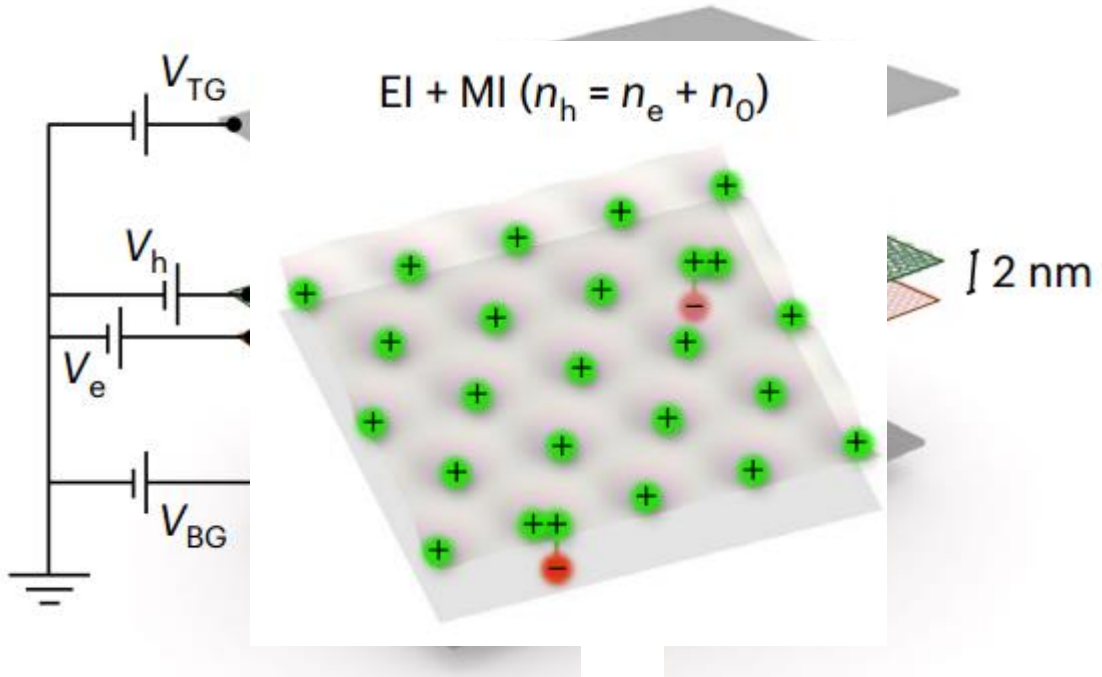
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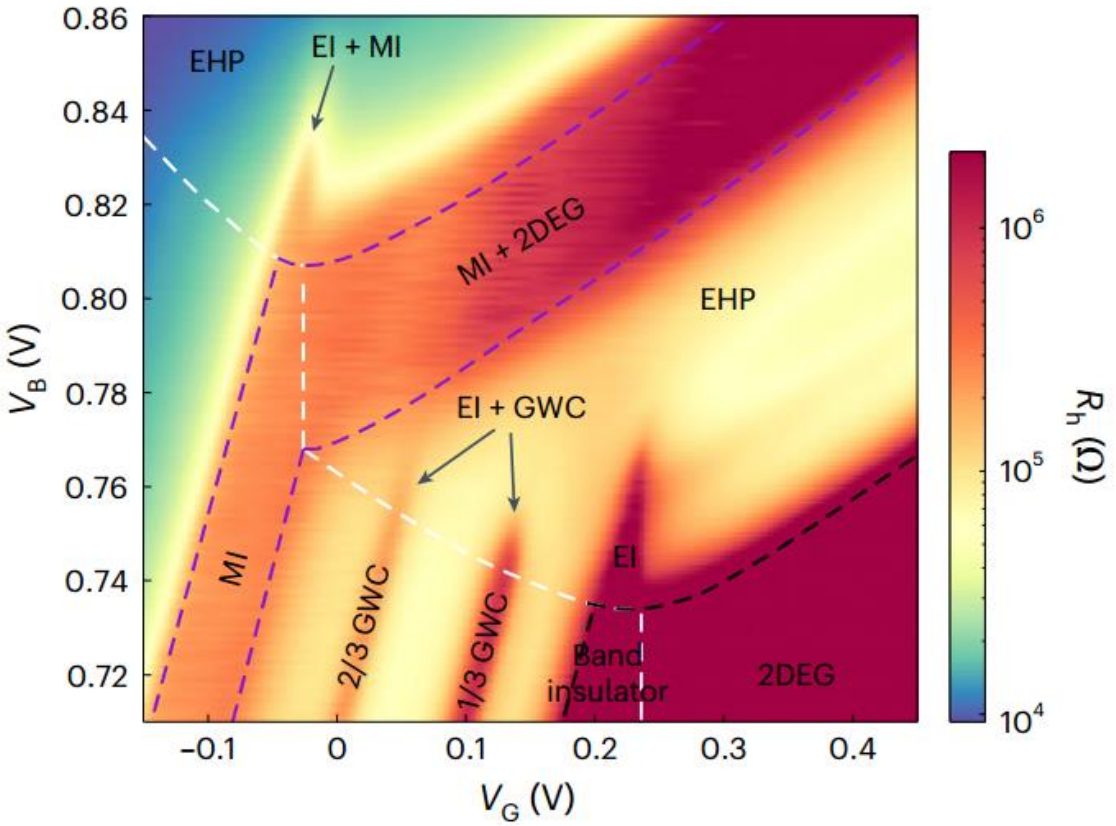


Nature Materials 23, 189–195 (2024)

In this system

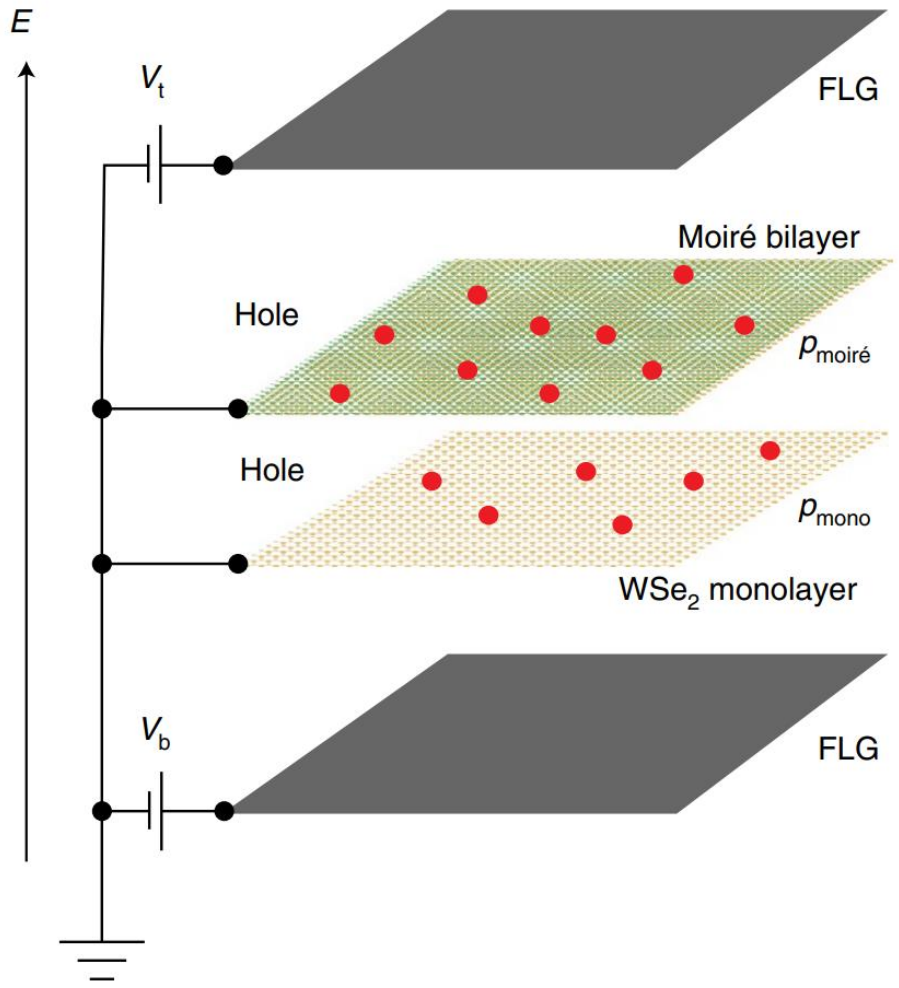
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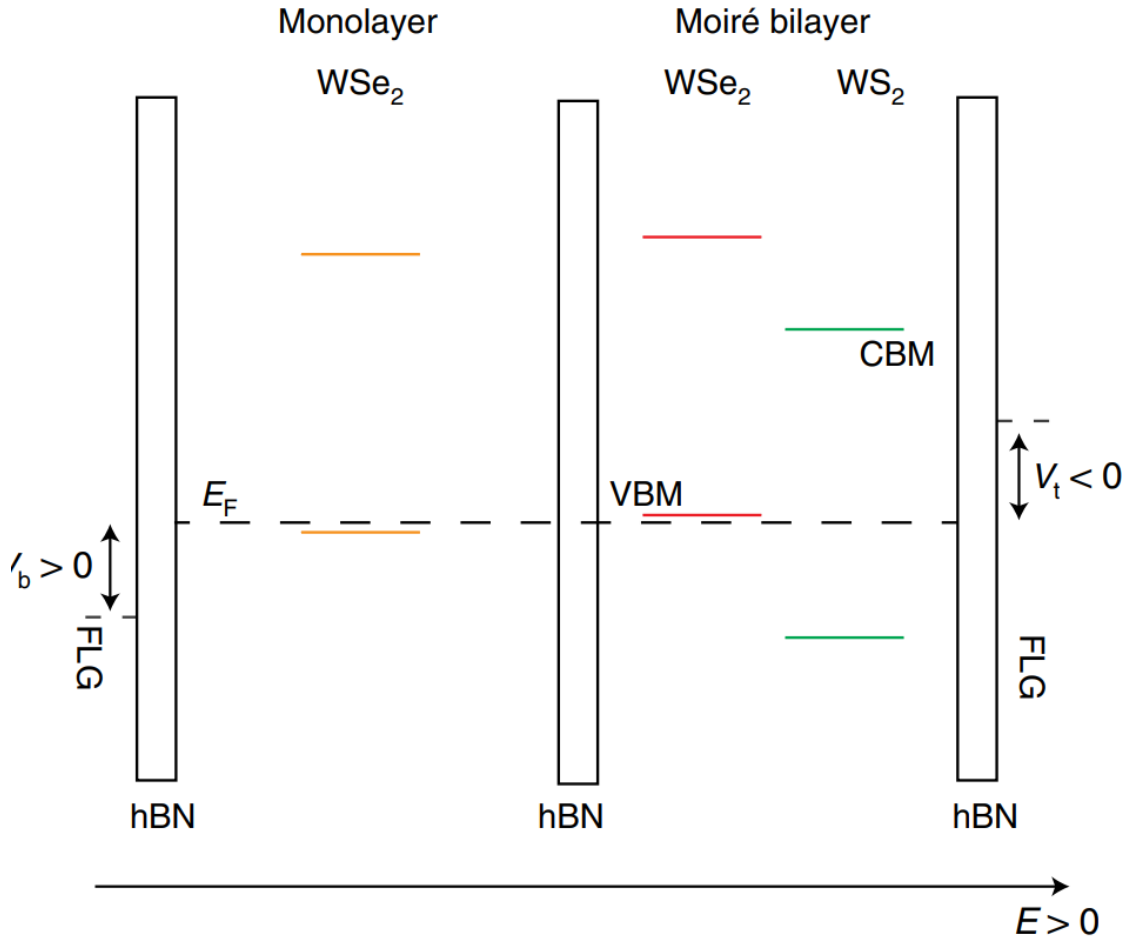




Nature Physics 22, 514–520 (2026)

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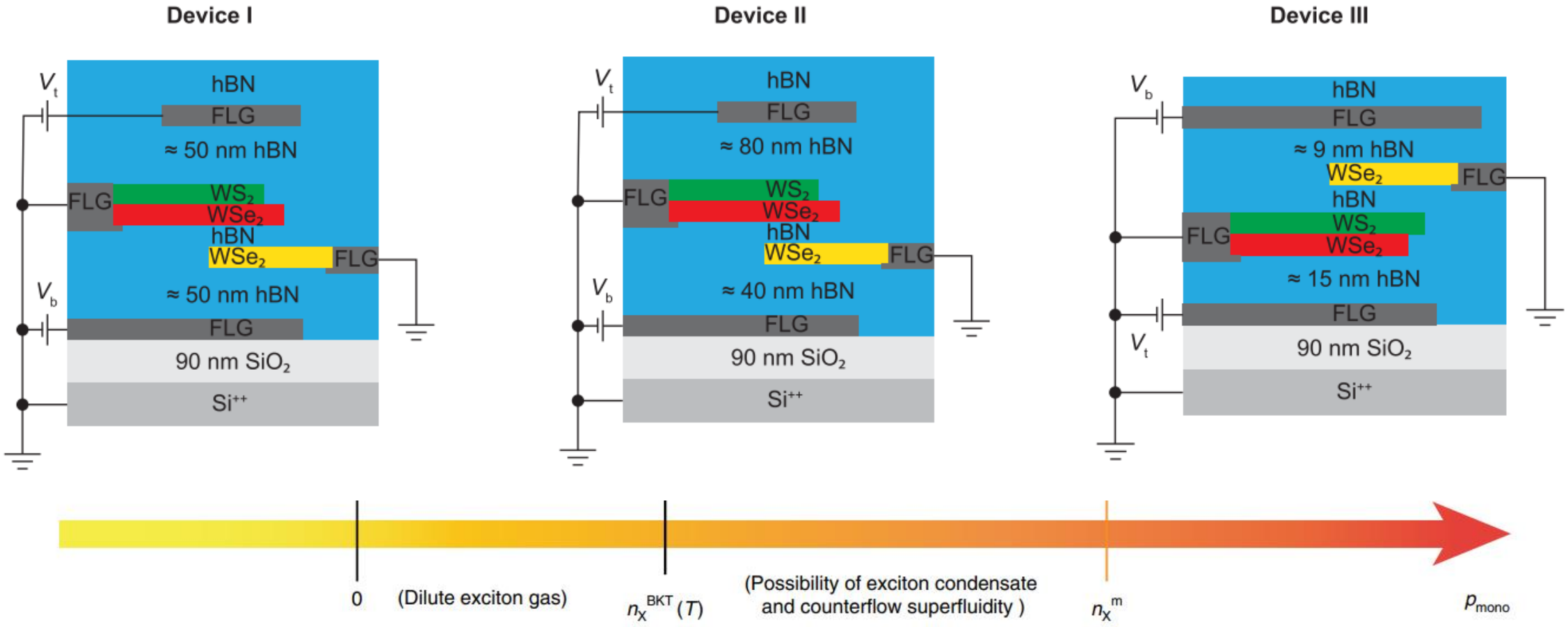




Nature Physics 18, 1214–1220 (2022)

In this system

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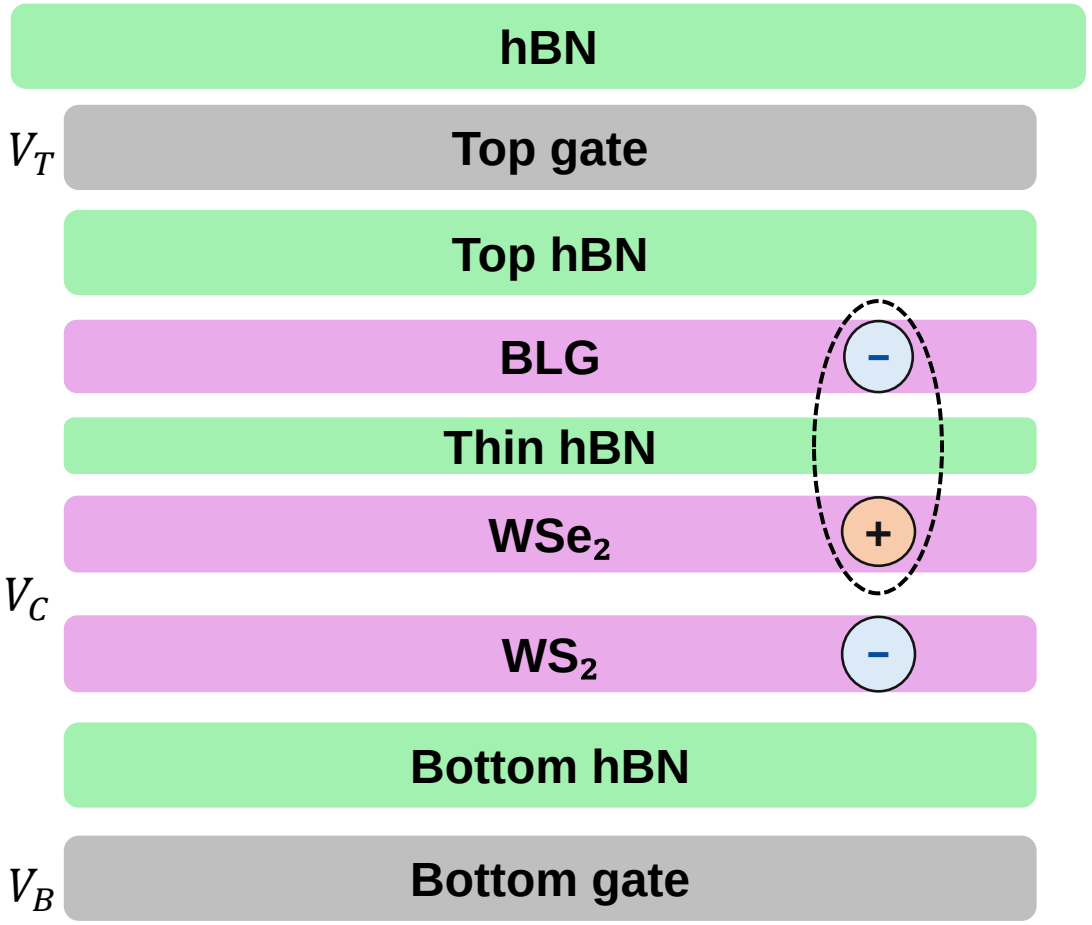
Nature Physics 18, 1214–1220 (2022)

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03.

Proposed Results

Tunability



Turn gate voltage to density/ field

With tunable magnetic field, we can change the kinetic energy dispersion and see the competition between interlayer binding and incompressible gaps.

Nature Physics 22, 514–520 (2026)

Proposed read-outs

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Define Bias gate V_b and symmetric gate V_g

In RC(E; V_b , V_g) spectrum map: 1s --- Typical doping TMDs;
2s --- Abnormal peak near $n_e \approx n_h$, attractive polaron appears and then disappears at incompressible regime of graphene. New translational order --- filling-selective Umklapp satellite

In RC(E; B, n_e) spectrum map:
AP/ RP branches appear near compressible Landau level region, being destroyed near graphene incompressible gap. Optical boundary move with B (with Rxx boundary)

In Rxx(V_b , V_g) resistance map:
Filling lines in different Fractional/Integer density ratio.
Rxx has diagonal insulating line which only appears at opening hole filling.

In Rxx(B, n_e) resistance map:
Normal graphene Landau-fan.
Around $l_B \sim a_M$ crossover appears and can be enhanced in partial Landau level.

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THANKS

Speaker: Zhongqi Xiu

In this system
Interesting properties?
Which is already been observed

What is unique about this artificial atom system
instead of interlayer exciton/ coloumb drag

Compared to feng's paper what's the advantage?
Tunable

How to observe the properties in hard way
(transport/RC) and confirm the property is unique

McCann & Koshino (2013); project calculation checklist

Helium teaches quantum statistics

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The analogy concerns collective behavior—not microscopic chemical composition.

^{4}He

bosonic quantum fluid

coherence possible when mobility
and phase stiffness are established

^{3}He

fermionic Fermi liquid

pairing is required before a
superfluid state can emerge

Composite parity may change with occupancy—but charge and internal states change too.

Kapitza (1938); Allen & Misener (1938); Leggett review